

Scratch Wound Assay Analysis Macro

Fiji / ImageJ Macro Documentation

Overview

This ImageJ/Fiji macro batch-processes a multi-scene transmitted light time-lapse acquired for a scratch wound (wound healing) assay. It has been developed for and tested on Zeiss Cell Discoverer 7 (CD7) data. For each selected scene ("series" / position in a multi-position acquisition), it segments the cell-free wound area in every frame of the time series, measures its size over time, and saves the resulting ROIs, an annotated overlay image, and a measurements table.

The macro relies on Bio-Formats for import, so it works with any Bio-Formats-readable multi-series file (e.g. Zeiss .czi, Nikon .nd2, Leica .lif, OME-TIFF, etc.) that contains multiple scenes/positions and a time dimension.

Input

Item	Description
Input file	A single microscopy file containing one or more scenes/series, each of which is a transmitted-light time series (T-stack). Selected via the file picker (input_file).
Output folder	An existing directory where all result files will be written (output_directory).

No other files are required. The macro reads image data directly from the input file via the Bio-Formats Importer for each requested scene.

script:Z:\Marie\Image_Analysis\Packages_plugins_and_scripts\Marie\work_in_progress\202... X

Please specify the files and parameters to be applied for the processing.

Select input file to be processed: Z:\path\time-series.czi

Select output folder to save result files to: Z:\path\analysis

First Scene to analyse: 1

Last Scene to analyse: 24

Scene increment for processing: 1

Median Filter radius (px): 2

Top Hat Filter radius (px): 30

Variance Filter radius (px): 35

Wound minimum size (micron): 100000

Adjustable Parameters

All parameters below are exposed as a Fiji Script Parameter dialog when the macro is run.

Parameter	Variable	Default	Description
First Scene to analyse	start_scene_index	1	Index of the first scene/series in the file to process.
Last Scene to analyse	end_scene_index	24	Index of the last scene/series to process.
Scene increment	scene_increment	1	Step size for looping through scenes (e.g. set to 2 to process every other scene).
Median Filter radius (px)	median_filter_radius	2	Radius of the median filter applied first, to reduce noise while preserving edges.
Top Hat Filter radius (px)	top_hat_filter_radius	30	Radius of the top-hat filter, used to even out uneven illumination/background and enhance contrast between the cell layer and the empty wound area.

Parameter	Variable	Default	Description
Variance Filter radius (px)	variance_filter_radius	35	Radius of the variance filter, which highlights textured (cell-covered) regions vs. the smoother, low-variance wound area.
Wound minimum size (μm^2)	wound_size_minimum	10000	Minimum particle area accepted as a "wound" region during particle analysis; smaller detected objects are discarded. Units follow the calibration of the input image (typically μm^2 for CD7 data).
Wound centroid allowed deviation from image centre (%)	wound_centroid_position_deviation_percent	50	Allowed region of the image that, if the ROI centroid lies outside the central, the ROI gets discarded and not drawn/measured. The percentage refers to percent of the whole image.

Additionally, two fixed (non-dialog) settings are configured at the start of the macro and are not user-adjustable without editing the code:

- Line width for ROI drawing/overlay: 6 px (run("Line Width...", "line=6")), reset to 1 px at the end.
- Measurements collected: area, mean, standard deviation, modal, min/max, centroid, center, perimeter, bounding box, fit ellipse, shape descriptors, Feret's diameter, integrated density, median, skewness, kurtosis, area fraction, stack position (run("Set Measurements...", ...)).

Processing Workflow

For each scene index from start_scene_index to end_scene_index (stepping by scene_increment):

1. Import

- The scene is loaded windowless via the Bio-Formats Importer as a hyperstack (series_list=<scene_index>).

2. Duplication

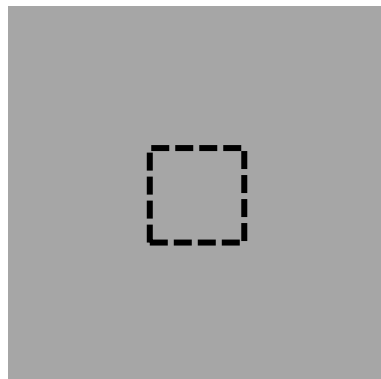
- The raw stack is duplicated twice: one copy (duplicate_ID) is kept as a clean reference for drawing the final wound outlines and for measurement; a second copy is used for segmentation processing.

3. Segmentation (applied per-frame across the whole stack)

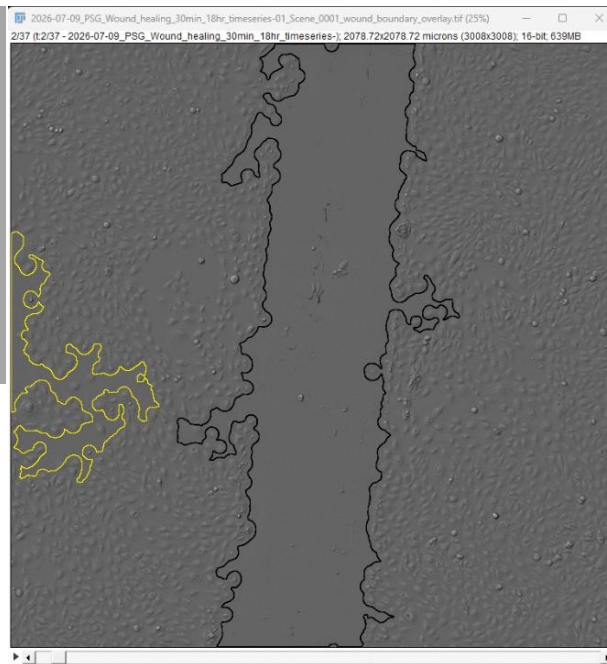
- Median filter (median_filter_radius) to denoise.
- Top Hat filter (top_hat_filter_radius) to correct uneven background/illumination.
- Variance filter (variance_filter_radius) to enhance the texture difference between the cell monolayer and the cell-free wound gap.
- Convert to Mask using the MaxEntropy auto-threshold method (dark background, calculated per stack, creates a binary mask stack).
- Fill Holes to close gaps within segmented regions.
- Invert so that the wound (originally the low-variance/dark region) becomes the foreground object to detect.

4. Particle analysis

- Analyze Particles is run on the mask stack with a size filter of wound_size_minimum-Infinity, generating a mask image and adding one ROI per detected wound region per frame to the ROI Manager.
- Any identified regions are filtered as follows:
 - If the ROI centroid lies outside the central region specified in percentage of the input image, the ROI gets discarded and not drawn/measured



Black outline: wound area
Yellow outline: area filtered out as the centroid position lies outside of the central 25% of the image (indicated in schema above)



5. Annotation & measurement

- For every ROI in the ROI Manager (i.e., every accepted wound detection in every frame): the ROI is measured on the clean duplicate stack, and its outline is drawn (black, 6 px line) directly onto the corresponding frame, producing a visual wound-boundary overlay burned into the image.

6. Saving results

- See the Output section below.

7. Cleanup

- The ROI Manager is reset, selections cleared, overlays hidden, and all open images are closed before moving to the next scene.

After all scenes are processed, the line width setting is reset to 1 px.

Output

For each processed scene, three files are written to the chosen output folder, named using the input file's base name and a zero-padded 4-digit scene number:

File	Content
<basename>_Scene_<0000>_LOG.txt	Console log with any progress output that was generated whilst each scene was processed.
<basename>_Scene_<0000>_ROIs-unfiltered.zip	ROI Manager archive containing one ROI per frame outlining the detected areas.
<basename>_Scene_<0000>_ROIs-filtered.zip	ROI Manager archive containing the filtered ROIs per frame outlining the detected wound area.
<basename>_Scene_<0000>_wound_boundary_overlay.tif	The original time-series image (as a TIFF stack) with the detected wound boundary drawn directly onto each frame, for visual QC.
<basename>_Scene_<0000>_wound_measurements.csv	Results table with one row per frame/ROI, containing area, shape descriptors, and other measurements (as configured by Set Measurements) describing the wound region over time.

Example, for input experiment1.czi, scene 3:

```
experiment1_Scene_0003_LOG.txt  
experiment1_Scene_0003_ROIs-filtered.zip  
experiment1_Scene_0003_ROIs-unfiltered.zip  
experiment1_Scene_0003_wound_boundary_overlay.tif  
experiment1_Scene_0003_wound_measurements.csv
```

No summary/combined file across scenes is generated — each scene produces its own independent set of three output files. If you need a combined dataset, the individual CSVs will need to be concatenated afterward. The CCI can give advice on this task.

Notes & Assumptions

- The macro assumes the wound region appears as a single, comparatively smooth/low-texture area relative to the surrounding cell monolayer in transmitted light images — this is what the top-hat + variance filtering combination is designed to exploit.
- `wound_size_minimum` should be adjusted based on image calibration (pixel size) and expected wound width; too low a value will pick up noise/debris as “wounds”, too high a value may fail to detect the wound in later frames as it closes.
- Using Centroid location as a filter can result in over filtering for sparse cell cultures but this also indicates that the wound has not been identified sufficiently. Open the relevant scene in Fiji and spot check the unfiltered ROI sets.
- If a scene does not contain a wound region above `wound_size_minimum` in a given frame, no ROI will be created for that frame, and it will simply be absent from the ROI set and CSV — gaps in the frame sequence should be interpreted as “no wound detected” for that frame, not as a healed wound (please spot-check the overlay TIFF).
- Processing time scales with the number of scenes and the size/length of each time series, since particle analysis is run per stack and drawing/measurement is done per detected ROI.
- Any pre-existing files in the output folder with the same generated names will be overwritten without warning.

Appendix

The Centre for Cell Imaging (CCI) facility supports imaging requirements of both existing users of the facility, as well as new potential users of the facility. Pilot studies are offered to check proof of principle for imaging experiments and are run by the CCI free of charge. The CCI is a non-profit shared facility with a transparent approach to cost recovery. Special discounted rates are available for University of Liverpool students, please contact [CCI@liv.ac.uk] for up-to-date charging structure.

The CCI facility aims to provide expert support for research as well as to provide training for researchers on the use of imaging techniques. The sustainability of the CCI facility relies on publications and acknowledgements resulting

from use of CCI equipment or expertise. To help with the sustainability of the CCI facility, please acknowledge the CCI as appropriate in any publication resulting from work obtained either through access to the equipment or expertise. For more information on how to acknowledge the facility, please refer to the CCI website [<https://cci.liv.ac.uk/furtherInformation.html>] and the Guidelines below.



Imaging Facility Guidelines for Acknowledgement

- 1** All publications resulting from the use of instruments and support staff within the facility should **acknowledge the facility as a whole**, e.g. 'the authors gratefully acknowledge the [core facility name] for their support & assistance in this work' and the facility should be **informed of the publication**.
- 2** Specific grants that have funded the facility instruments used for the work to be published must be acknowledged if the data was acquired during the active period of that grant. **Facility staff will advise users of such grant codes**.
- 3** Assistance provided above the technical or routine level, with any facility staff providing scientific input and expertise in experimental set-up, acquisition, analysis or writing, should be recognised through **co-authorship** on resulting publications. Please discuss authorship and acknowledgement with facility staff prior to manuscript submission.

Example scenarios:

Sample Preparation	Routine sample preparation following standard protocol.	Simple acknowledgement
	Development of new sample preparation protocols. Optimisation of existing protocols for specific samples.	Inclusion of specific facility member on author list
Image Acquisition	Training of users to acquire images themselves. Simple acquisition of raw data.	Simple acknowledgement
	Operational image acquisition with input and decisions dependent on expertise. Design or re-design of experimental conditions.	Inclusion of specific facility member on author list
Image Analysis	Recommendation of analysis software and tools. Basic data analysis help and advice.	Simple acknowledgement
	Constructive data analysis and interpretation. Creation of complex custom image analysis tools.	Inclusion of specific facility member on author list

Based on the publication policy compiled by Natasha Stephen, Plymouth Electron Microscopy Centre, after discussions with the RMS EM-UK community.

